

Introduction

Definition of Project Management

Project management in the modern sense began in the early 1950s, although it has its roots much further back in the latter years of the 19th century. The need for project management was driven by businesses that realised the benefits of organising work around projects and the critical need to communicate and co-ordinate work across departments and professions. One of the first major uses of project management as we know it today was to manage the US space programme.

Project management is the discipline of planning, organizing, securing, managing, leading, and controlling of resources to achieve specific goals.

The Project Management Institute's (2004) define project management is the application of knowledge, skills, tools and techniques to project activities to meet project requirements.

According to **Chase and Aquilano**, Project Management can be defined as the planning, directing and controlling of resources (people, equipment, material) to meet technical, cost and time constraints of the project.

Project management is often summarised in a triangle (see Figure 1). The three most important factors are time, cost and scope, commonly called the triple constraint. These form the vertices with quality as a central theme



Figure 1: The triple constraint

- Projects must be within cost.
- Projects must be delivered on time.
- Projects must be within scope.
- Projects must meet customer quality requirements.

More recently, this has given way to a project management diamond, with cost, time, scope and quality the four vertices and customer expectations as a central theme (see Figure 2). No two customer expectations are the same so you must ask what their expectations are.



Figure 2: The project management diamond

Functions of Project Management

No matter what the type of project, project management typically follows the same pattern:

1. Defining the Project

In this stage the project manager defines what the project is and what the users hope to achieve by undertaking the project. This phase also includes a list of project deliverables, the outcome of a specific set of activities. The project manager works with the business sponsor or manager who wants to have the project implemented and other stakeholders -- those who have a vested interest in the outcome of the project.

2. Planning the Project

Define all project activities. In this stage, the project manager lists all activities or tasks, how the tasks are related, how long each task will take, and how each task is tied to a specific deadline. This phase also allows the project manager to define relationships between tasks, so that, for example, if one task is x number of days late, the project tasks related to it will also reflect a comparable delay. Likewise, the project manager can set milestones, dates by which important aspects of the project need to be met.

3. Executing the Project

Build the project team. In this phase, the project manager knows how many resources and how much budget he or she has to work with for the project. The project manager then assigns those resources and allocates budget to various tasks in the project. Now the work of the project begins.

4. Controlling the Project

The project manager is in charge of updating the project plans to reflect actual time elapsed for each task. By keeping up with the details of progress, the project manager is able to understand how well the project is progressing overall.

5. Closure of the Project

In this stage, the project manager and business owner pull together the project team and those who have an interest in the outcome of the project (stakeholders) to analyze the final outcome of the project.

Importance of Project Management

Project management is a crucial factor in determining the success of an organization. It plays a vital role in achieving growth, handling economic challenges, and increasing overall performance.

1. Ensures High Product Quality

Proper planning, budgeting, and quality control help organizations produce high-quality products, enhancing customer satisfaction and goodwill.

2. Promotes Effective Communication

Project managers act as a link between team members and stakeholders, ensuring clear communication, better coordination, and timely updates.

3. Reduces Risks

Project management helps identify potential risks early and take preventive actions, saving time and resources.

4. Supports Strategic Goals

It aligns project activities with organizational objectives and ensures efficient use of resources, leading to better decision-making and innovation.

5. Improves Crisis Management

Project managers handle unexpected challenges such as financial issues or natural disasters through proper planning and quick decision-making.

6. Enhances Resource Utilization

Project management ensures the optimal use of available resources, including manpower, materials, and finances. It prevents wastage and improves productivity, which ultimately increases profitability.

7. Ensures Timely Completion of Projects

Meeting deadlines is critical for organizational success. Project management uses tools like scheduling, Gantt charts, and performance tracking to ensure that projects are completed within the specified time frame.

Project management is essential for all organizations, as it ensures timely completion, efficient resource use, and improved performance, helping firms achieve long-term success.

PROJECT FORMULATION

Concept of Project Formulation

Project formulation is the systematic development of a project idea for the eventual objective of arriving at an investment decision. It involves a step-by-step investigation and development of project idea. And it provides a controlled mechanism for restricting expenditure on project development.

According to **B.B. Goel**, Project formulation refers to a series of steps to be taken to convert an idea or aspiration into a feasible plan of action.

According **G. Myrdal**, Project formulation is one of the basic techniques through which planning can be changed from an idea to an institutional and rational base.

Project formulation is a process involving the joint efforts of a team of experts. Each member of the team should be familiar with the broad strategy, objectives and other ingredients of the project. The government official who deals with the project's final clearance has to be treated as forming part of the team. A well-formulated feasibility report provides a medium, which cuts across scientific, social and positional prejudices and provides a common meeting ground for all those who have a contribution to make in successful implementation of a project.

Project team should consist of experts in major substantive fields of the project. Depending on the situation any large project should comprise the following team members.

- a. One industrial economist
- b. One market analyst
- c. One or more technologist/engineer specializing in the appropriate industry.
- d. One mechanical and/or industrial engineer.
- e. One management accounting expert.

Stages of Project Formulation

1. Project identification

Project identification is concerned with the collection, compilation and analysis of economic data for the eventual purpose of locating possible opportunities for investment and with the development of the characteristics of such opportunities.

2. Technical Analysis

Technical analysis includes the following:

1. Input analysis

a. Site: Location of enterprise; owned or leasehold land; industrial area; No Objection Certificate from the Municipal Authorities if the enterprise location falls in the residential area.

- b. Raw Material:** Requirement of raw material, whether inland or imported, sources of raw material supply.
- c. Human Resource:** Availability of skilled, Semi-skilled labor in the area, arrangements for training laborers in various skills.
- d. Technology Selected:** The selection of technology, arrangements made for acquiring it should be mentioned in the business plan.
- e. Utilities:** These include: **Power:** Requirement for power, load sanctioned, availability of power. **Fuel:** Requirement for fuel items such as coal, coke, oil or gas, state of their availability. **Water:** The sources and quality of water should be clearly stated in the project report.
- f. Pollution Control:** The aspects like scope of dumps; sewage system and sewage treatment plant should be clearly stated in case of industries producing emissions.
- g. Communication System:** Availability of communication facilities, e.g., telephone, fax, internet facility, etc. should be stated in the project report.
- h. Transport Facilities:** Requirements for transport, mode of transport, potential means of transport, distances to be covered, bottlenecks etc., should be stated in the business plan.
- i. Other Common Facilities:** Availability of common facilities like machine shops, welding shops and electrical repair shops etc. should be stated in the report.
- j. Production Process:** A mention should be made for process involved in production and period of conversion from raw material into finished goods.
- k. Machinery and Equipment:** A complete list of items of machinery and equipments required indicating their size, type, cost and sources of their supply should be enclosed with the project report.
- l. Capacity of the Plant:** The installed licensed capacity of the plant along with the shifts should also be mentioned in the project report.
- m. Research and Development:** A mention should be made in the project report regarding proposed research and development activities to be undertaken in future.

2. Demand and Supply analysis

State the total expected demand for the product and present supply position. This should also be mentioned how much of the gap will be filled up by the proposed unit. We can determine the demand of consumer's in following way

- a. Survey of buyer intentions;
- b. Composite of sales force opinion;
- c. Expert's opinion;
- d. Market test method;
- e. Statistical method.

3. Feasibility study

Feasibility studies aim to objectively and rationally uncover the strengths and weaknesses of an existing business or proposed venture, opportunities and threats as presented by the environment, the resources required to carry through, and ultimately the prospects for success.

4. Project Appraisal

Project appraisal is the systematic and comprehensive review of the economic, environmental, financial, social, technical and other such aspects of a project to determine if it will meet its objectives.

According to **B B Goel**, Project appraisal is a tool to examine as to whether in the given situation, it would be most realistic, reliable and reasonable one to commit resources or not.

According to **P.K. Motto**, Project appraisal is the process of evaluation the salient features of feasibility analysis, techno-economic analysis, design and network analysis, input analysis, financial analysis and social cost-benefit analysis of a project.

The following specific aspects must be covered

- Technical Viability
- Commercial Viability - Market Demand
- Financial Viability
- Economic Viability
- Environmental Viability
- Managerial Viability

In essence, the project should be technically feasible and adequate demand should exist for selling the products generated by the operation of the project facilities. There should be adequate financial returns, and adequate returns to the national economy. All environmental requirements must be possible to be fulfilled and the enterprise must have the capability to manage the facilities efficiently and effectively after the project is over.

PROJECT PLAN

Definition of Project Plan

Project planning is a procedural step in project management, where required documentation is created to ensure successful project completion. Documentation includes all actions required to define, prepare, integrate and coordinate additional plans. The project plan clearly defines how the project is executed, monitored, controlled and closed.

A Project Plan sets out the phases, activities and tasks needed to deliver a project. The timeframes required to deliver the project, along with the resources and milestones are also shown in the Project Plan.

According to the **Project Management Body of Knowledge**, Project plan is a formal, approved document used to guide both project execution and project control. The primary uses of the project plan are to document planning assumptions and decisions, facilitate communication among stakeholders, and document approved scope, cost, and schedule baselines. A project plan may be summarized or detailed.

According to **R L Martino**, A project plan establishes the duration of the project, the resources needed to complete each activity and the required sequence of performance of each job.

Project planning requires an in-depth analysis and structuring of the following activities:

- Setting project goals
- Identifying project deliverables
- Creating project schedules
- Creating supporting plans

Steps of Project Planning

Project planning is the most important part of project management, as it provides direction for successful execution. The key steps are:

1. Project Goals

Identify all stakeholders such as sponsors, customers, users, and the project team. Gather their needs through interviews, prioritize them, and convert them into clear, measurable (SMART) goals.

2. Project Deliverables

Based on the goals, list all outputs the project must produce. Define what will be delivered, when, and in what form, with estimated deadlines.

3. Project Schedule

Break deliverables into tasks and estimate time, effort, and required resources. Assign responsibilities and set realistic deadlines. If deadlines are unrealistic, adjust scope, resources, or time.

4. Supporting Plans

a. Human Resource Plan

Define team members, their roles, responsibilities, and required manpower.

b. Communication Plan

Decide how project information will be shared with stakeholders (e.g., reports, meetings, updates).

c. Risk Management Plan

Identify potential risks, prepare preventive actions, and create contingency plans using a risk log.

A well-structured project plan ensures clear goals, efficient resource use, proper communication, and risk control, leading to successful project completion.

Importance of Project Planning

Project planning is essential before starting any project because it provides direction and helps ensure successful completion within time, cost, and quality limits. Without proper planning, projects are more likely to fail due to confusion and lack of structure.

1. Reducing Risk

Planning helps identify potential risks early, allowing managers to avoid or minimize them before they become serious problems.

2. Reducing Uncertainty

Planning improves understanding of the project, technology, and team capabilities, helping to refine goals and avoid developing the wrong product.

3. Better Decision Making

Accurate estimates of cost, time, and resources help organizations decide whether a project is worthwhile and which tasks should be prioritized.

4. Building Trust

Reliable planning and delivery increase trust between project teams and clients, as stakeholders can depend on realistic expectations and timelines.

5. Effective Communication

A project plan clearly communicates goals, expectations, and timelines, helping all stakeholders understand what will be delivered.

Project planning is crucial for reducing risk, improving decisions, and ensuring smooth execution. A flexible and well-structured plan greatly increases the chances of project success.

PROJECT SCHEDULE

📁 Definition of Scheduling

In project management, a schedule consists of a list of a project's terminal elements with intended start and finish dates. Terminal elements are the lowest element in a schedule, which is not further subdivided. Those items are often estimated in terms of resource requirements, budget and duration, linked by dependencies and scheduled events.

A schedule is the conversion of a project action plan an operating timetable. It serves as the basic for monitoring and controlling project activity and taken together with the plan and budget.

According to **R L Martino**, A schedule depicts the expected start and finish time of each job. It is produced by allocating resources upto the limit of availability according to the requirements given by the plan.

Scheduling is the determination of time required for execution of each operation and the time order in which each operation has to be carried out, to meet the plan objectives.

📁 Network Techniques (Critical Path Method)

Network techniques Developed in 1950's CPM by DuPont for chemical plants (1957)

According to **James E. Kelley and Morgan R. Walker**, network techniques (CPM) are defined as a method of planning, scheduling, and controlling projects by representing activities in a network diagram to determine the longest sequence of dependent tasks (critical path) that governs the total project duration.

As stated by **Keith Lockyer**, network techniques are systematic methods of planning and controlling projects by breaking them into activities and showing their sequence and interrelationships in a network diagram to improve coordination and decision-making.

According to **Jay Heizer and Barry Render**, network techniques are analytical tools used to schedule, monitor, and control complex projects by identifying task dependencies and determining the critical path for efficient project completion.

CPM models the activities and events of a project as a network. Activities are shown as nodes on the network and events that signify the beginning or ending of activities are shown as arcs or lines between the nodes. The Figure shows an example of a CPM network diagram:

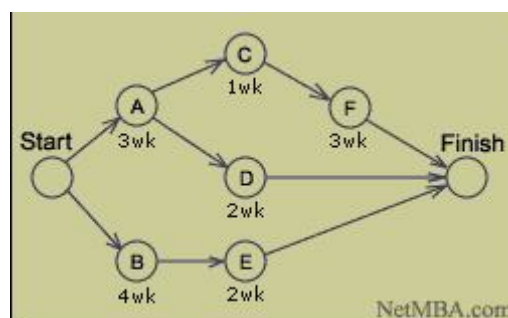
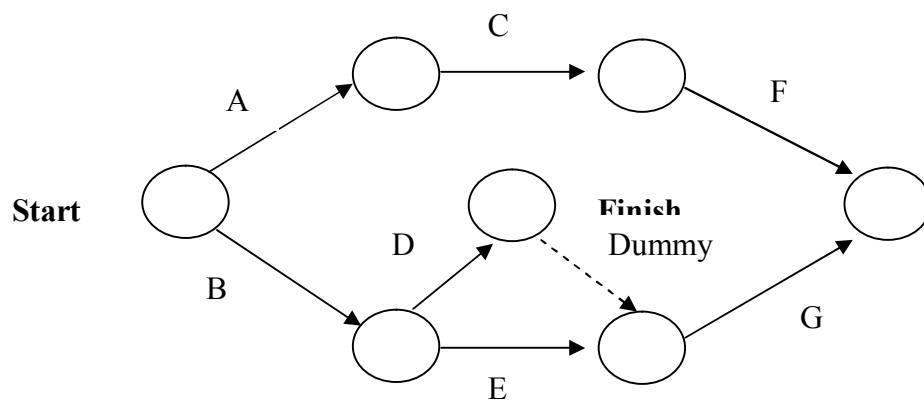
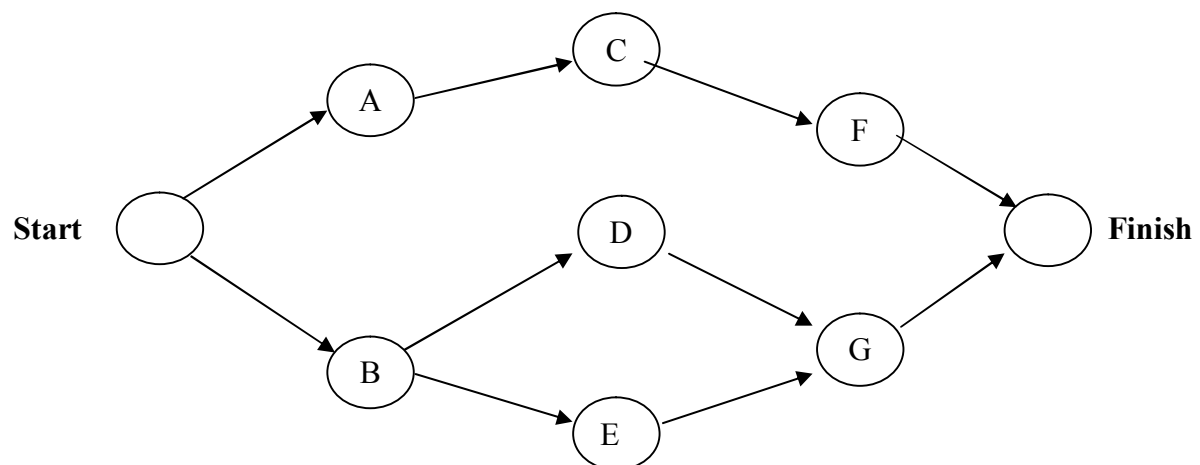


Figure: CPM Diagram

Arrow and Node diagram

Activity	Immediate predecessor activity
A	-
B	-
C	A
D	B
E	B
F	C
G	D, E

**Figure: Arrow Diagram****Figure: Node Diagram**

Following are the information of a construction project, using the given information- Draw a network diagram and determine critical path.

Activity	Time of work (in week)	Immediate predecessor activity
A	6	-
B	9	A
C	8	A
D	12	A
E	11	B
F	7	C, D
G	12	E, F
H	14	D
I	9	G, H

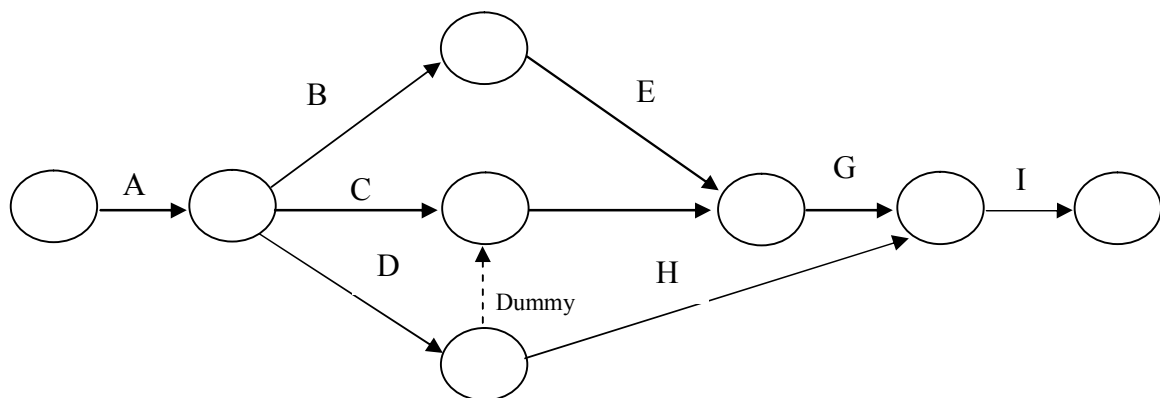


Figure: Network Diagram

Determination of Critical path

If we want to determine the critical path, we have to find out the alternative paths from the above diagram and determine their length. We have found out the following alternative paths from the above diagram:

1. A – B – E – G – I = 6+9+11+12+9 = 47 weeks
2. A – C – F – G – I = 6+8+7+12+9 = 42 weeks
3. A – D – Dummy – F – G – I = 6+12+0+7+12+9 = 46 weeks
4. A – D – H – I = 6+12+14+9 = 41 weeks

We know that, among all the alternative paths, longest path is determined as Critical Path. A-B-E-G-I path within the above mentioned diagram is the longest path among all the alternative paths. Hence, Critical path = 47 weeks.

PROJECT IMPLEMENTATION

Concept of Project Implementation

Implementation is the stage where all the planned activities are put into action. Before the implementation of a project, the implementors (spearheaded by the project committee or executive) should identify their strength and weaknesses (internal forces), opportunities and threats (external forces). The strength and opportunities are positive forces that should be exploited to efficiently implement a project. The weaknesses and threats are hindrances that can hamper project implementation. The implementors should ensure that they devise means of overcoming them.

According to **Harold Kerzner**, project implementation is the stage where “project plans are put into action and project work is carried out to achieve desired results.”

According to **Jack R. Meredith & Samuel J. Mantel Jr.**, project implementation is the “process of implementing the project plan and coordinating people and resources to achieve intended results.”

According to **Dennis Lock**, project implementation refers to “the stage where all planned activities are executed, monitored, and controlled to achieve the project objectives.”

Implementation is the carrying out, execution or practice of a plan, a method or any design for doing something. As such, implementation is the action that must follow any preliminary thinking in order for something to actually happen.

Steps of project implementation

Project implementation is a structured process that ensures effective communication, coordination, and successful project completion. The main steps are:

1. Seek Guidance

A project manager should get advice from experienced professionals or mentors, especially if they are new to project management.

2. Acquire Tools and Software

Proper tools such as communication systems, computers, and project management software should be arranged to support smooth implementation.

3. Clarify Project Details

The project plan should clearly define objectives, timeline, budget, team members, methods, and risk management strategies.

4. Assign Responsibilities

Roles and duties of all team members should be clearly defined, and proper communication channels must be established.

5. Monitor Progress

Regular tracking of project performance, quality, time, and cost should be done to ensure the project stays on schedule.

6. Document Information

All project activities, progress reports, decisions, and problems should be properly recorded for future reference and control.

7. Communicate with Stakeholders

Continuous communication with upper management and clients is necessary to keep them updated on project progress.

Conclusion

Following these steps helps ensure effective project implementation and increases the chances of project success.

Impediments to Project Implementation and Strategies to Overcome Them

Successful project implementation often faces several obstacles that can delay progress, increase costs, or reduce quality. These impediments usually arise from managerial, technical, financial, and environmental factors. Understanding them and applying suitable strategies is essential for effective project completion.

1. Poor Planning and Unclear Objectives

One of the major problems in project implementation is weak planning or unclear project goals. When objectives are not properly defined, confusion arises among team members, leading to inefficiency and delays.

Strategy:

Proper project planning using tools like Work Breakdown Structure (WBS), clear goal setting (SMART objectives), and detailed scheduling can help ensure clarity and direction from the beginning.

2. Lack of Adequate Resources

Projects often fail due to insufficient human resources, financial constraints, or lack of required materials and technology.

Strategy:

Conduct proper resource planning during the initiation phase, ensure realistic budgeting, and allocate skilled personnel effectively. Resource leveling techniques can also help balance workload.

3. Poor Communication

Ineffective communication between project managers, team members, and stakeholders can lead to misunderstandings, errors, and delays.

Strategy:

Establish a strong communication plan, regular meetings, progress reports, and use of project management software to ensure transparency and coordination.

4. Risk and Uncertainty

Unexpected risks such as market changes, technical failures, political instability, or natural disasters can disrupt project execution.

Strategy:

Implement risk management practices such as risk identification, risk assessment, and mitigation planning. Contingency plans should always be prepared in advance.

5. Lack of Skilled Workforce

A project may suffer if the team lacks the necessary technical or managerial skills.

Strategy:

Invest in training and development programs, hire experienced professionals, and provide continuous skill enhancement opportunities.